



SYMBIOSIS INSTITUTE OF TECHNOLOGY (SIT)

Constituent of Symbiosis International (Deemed University), Pune

(Established under Section 3 of the UGC Act of 1956 vide notification number F-9-12/2001-U-3 of the Government of India)
Re-Accredited by NAAC with 'A++' Grade

Subject: Programming with Java

ASSIGNMENT No : 5

TITLE : Develop a Java application demonstrating the use of ArrayList, Vector, and StringBuffer for dynamic data storage, management, and string manipulation.

Problem Statement

Develop a Java application demonstrating the use of ArrayList, Vector, and StringBuffer for dynamic data storage, management, and string manipulation.

Learning Objectives

To implement dynamic data storage and string manipulation using ArrayList, Vector, and StringBuffer.

Theory

ArrayList and Vector are dynamic array implementations provided by the Java Collections Framework. ArrayList offers faster performance in single-threaded applications, whereas Vector provides synchronized operations for thread safety. StringBuffer is a mutable sequence of characters that allows efficient string modification without creating new objects. These classes are commonly used in applications requiring dynamic memory allocation and efficient string processing.

Outcome

Exercise

1. Create a **To-Do List application** using ArrayList to store tasks and StringBuffer to display tasks.

2. Create a **Student Course Registration System** using ArrayList to store the list of courses registered by a student and **StringBuffer** to generate and display the registered course list.

The application should allow users to add, remove, and view registered courses.

CODE

Users > raghav_oze > Desktop > java > exp5 >  todolist.java >  todolist >  main(String[] args)

```
1  import java.util.ArrayList;
2  class todolist
3  {
4      Run main | Debug main
5      public static void main(String args[])
6      {
7          ArrayList<String> list = new ArrayList<>();
8          StringBuffer task = new StringBuffer();
9          task.append("complete assignments");
10         list.add(task.toString());
11         System.out.println("\nTo-Do List:");
12         for (String item : list)
13         {
14             System.out.println("- " + item);
15         }
16         list.add("go to gym");
17         list.add("sleep 7hours");
18         System.out.println("\nTo-Do List:");
19         for (String item : list)
20         {
21             System.out.println("- " + item);
22         }
23         list.remove(0);
24         System.out.println("\nTo-Do List:");
25         for (String item : list)
26         {
27             System.out.println("- " + item);
28         }
29     }
30 }
31
32
```

Output

```
zsh: command not found: arraylist.class
● raghav_oze@raghavs-MacBook-Air-3 exp5 % java todolist.java

To-Do List:
- complete assignments

To-Do List:
- complete assignments
- go to gym
- sleep 7hours

To-Do List:
- go to gym
- sleep 7hours
❖ raghav_oze@raghavs-MacBook-Air-3 exp5 %
```

CODE

```
J convert.java U ● J todoist.java 1, U J courses.java 1, U ●
Users > raghav_oze > Desktop > java > exp5 > J courses.java > courses > main(String[] args)
1  /*Create a Student Course Registration System using ArrayList to store the list of courses
2  registered by a student and StringBuffer to generate and display the registered course list.
3  The application should allow users to add, remove, and view registered courses. */
4
5  import java.util.ArrayList;
6  class courses
7  {
8      Run main | Debug main
      public static void main(String args[])
9      {
10         ArrayList<String> courseList = new ArrayList<>();
11         StringBuffer registeredCourses = new StringBuffer();
12         courseList.add("Mathematics");
13         courseList.add("Physics");
14         courseList.add("Chemistry");
15         System.out.println("\nRegistered Courses:");
16         for (String course : courseList)
17         {
18             registeredCourses.append(course).append(", ");
19         }
20         System.out.println(registeredCourses.toString());
21         courseList.add("Biology");
22         System.out.println("\nRegistered Courses");
23         registeredCourses.setLength(0);
24         for (String course : courseList)
25         {
26             registeredCourses.append(course).append(", ");
27         }
28         System.out.println(registeredCourses.toString());
29         courseList.remove("Physics");
30         System.out.println("\nRegistered Courses ");
31         registeredCourses.setLength(0);
32         for (String course : courseList)
33         {
34             registeredCourses.append(course).append(", ");
35         }
36         System.out.println(registeredCourses.toString());
37     }
38 }
```

OUTPUT

```
⊗ raghav_oze@raghavs-MacBook-Air-3 exp5 % courses.java
zsh: command not found: courses.java
● raghav_oze@raghavs-MacBook-Air-3 exp5 % java courses.java

Registered Courses:
Mathematics, Physics, Chemistry,

Registered Courses after adding Biology:
Mathematics, Physics, Chemistry, Biology,

Registered Courses after removing Physics:
Mathematics, Chemistry, Biology,
❖ raghav_oze@raghavs-MacBook-Air-3 exp5 %
```